Overall, while the IAT has clear methodological limitations, it remains a useful tool for encouraging self reflection and discussion about bias, especially within the context of data ethics and responsible technology development.

Thoughts on criticism

In my view, the Implicit Association Test is useful as a tool for raising awareness rather than as a definitive measure of individual bias. The exercise encouraged reflection on how unconscious associations can exist even when people consciously reject biased views. However, I am cautious about how the results are interpreted, particularly when simplified diagnostic labels are presented to the general public.

I agree with criticisms that the IAT may capture factors such as cultural exposure, familiarity, or task related effects rather than stable personal attitudes. The fact that results can vary across repeated attempts and be influenced by the order of category pairings suggests that the test lacks the consistency expected of a diagnostic instrument. For this reason, I do not believe it should be used to draw firm conclusions about an individual's character or intentions.

That said, I do not think these limitations make the IAT meaningless. When used responsibly, it highlights how bias can operate indirectly and unintentionally, which is particularly relevant in areas such as data collection, annotation and algorithm design. Overall, I see the IAT as a starting point for discussion and ethical reflection rather than a precise measurement of personal bias. This perspective aligns with the need for caution and transparency when dealing with bias in both human judgement and automated systems.

Practitcal 3

Task 1: What are the AI principles and concerns relevant to your group work topic?
Produce a short summary of 300- 500 words. Include sub-headings to structure the prose and highlight key content. Try to exemplify issues with recent examples if possible. Using Harvard referencing, cite a blend of relevant sources, academic articles, quality news items and other key documents or sources. List the full citations details for the sources, in alphabetical order, in a References section at the end.

Intro

Artificial intelligence is increasingly embedded within transport systems, including traffic optimisation, autonomous vehicles, predictive maintenance and public transport management. While these applications promise efficiency and safety improvements, they also raise ethical concerns that must be addressed through Responsible AI principles.

Fairness and non discrimination

Fairness is a central concern in AI-driven transport systems. Algorithms trained on historical traffic or mobility data may reflect existing social and geographic inequalities, such as disproportionate surveillance or enforcement in certain neighbourhoods. Research has shown that algorithmic decision systems can reinforce structural bias if fairness is not explicitly addressed during design and deployment (Fjeld et al., 2020). In transport contexts, this may lead to unequal access to services or biased traffic enforcement outcomes.

Transparency and explainability

Transparency is essential for public trust in AI systems that operate in shared physical spaces. Many transport-related AI models function as black boxes, making it difficult for users or regulators to understand how decisions are made. The European Commission (2019) emphasises that explainability is particularly important where AI affects individuals directly, such as congestion pricing, route prioritisation or autonomous vehicle behaviour. Clear explanations of system logic and limitations support accountability and informed public oversight.

Accountability and human oversight

Responsible AI frameworks stress that humans must remain accountable for AI system outcomes. In safety-critical transport applications, such as autonomous driving or rail signalling, meaningful human oversight is essential to manage failures or unexpected behaviour (Floridi et al., 2018). Without clear governance structures, responsibility can be diffused between developers, operators and systems, increasing the risk of harm.

Safety, robustness and privacy

AI transport systems must operate reliably across varied conditions, including adverse weather and sensor failure. Robust testing reduces the risk of accidents caused by system

error (ENISA, 2023). In addition, many transport applications rely on sensitive location and mobility data. Responsible AI requires strong data protection practices to prevent misuse and maintain public trust.

Task 2: Consider how dark patterns relate (or could relate) to your topic.
Produce a short summary of 300-500 words styled in the same way as the above summary.
3. For both tasks, reflect generally on your findings and on the performance of the AI outputs and any tell-tale signs that would suggest AI had been used. In particular, reflect on the quality and relevance of selected sources and their citation details.

Dark patterns in AI based transport systems

Dark patterns are design practices that manipulate user behaviour, often undermining autonomy or informed consent. In AI-based transport systems, dark patterns can emerge in interfaces, data collection practices and automated decision feedback.

Manipulative Consent and Privacy Dark Patterns

AI transport services frequently rely on continuous location tracking and behavioural data. Dark patterns such as default opt-in settings or vague consent language can pressure users into sharing more data than intended. Research on dark patterns highlights how interface design can exploit user trust and cognitive shortcuts, particularly in digital services framed as essential (Gray et al., 2018).

Roach motel and restricted user control

Another relevant dark pattern is the roach motel, where users can easily consent to data collection but face difficulty withdrawing consent. In transport applications, opt-out mechanisms for AI-driven tracking or data retention may be hidden within complex settings menus. This undermines transparency and conflicts with Responsible AI principles of autonomy and control (Mathur et al., 2019).

Manipulative defaults and automated feedback

AI systems may also influence transport decisions through manipulative defaults, such as prioritising toll roads, commercial partners or data-rich routes without clear justification.

When automated outputs are presented as neutral or optimal, users may be discouraged from questioning decisions. This can obscure accountability and reduce opportunities for contesting biased or incorrect outcomes.

Ethical implications

Dark patterns in transport systems risk eroding public trust and amplifying power imbalances between service providers and users. Addressing these risks requires ethical interface design, clear consent mechanisms and regulatory oversight to ensure AI systems support user interests rather than exploit behavioural tendencies.

Task 3: Reflection on AI outputs, sources and signs

AI tools can be helpful for structuring content and identifying relevant themes in Responsible AI and dark pattern discussions. However, their outputs often require careful human review. AI generated text may appear confident and well-structured while lacking sufficient critical depth or contextual nuance.

A key limitation is source reliability. AI tools may suggest references that appear plausible but are outdated, non peer reviewed or inaccurately cited. This exercise highlighted the importance of verifying sources independently and prioritising peer-reviewed research, official policy documents and reputable academic publications. Human judgement was necessary to ensure that all citations were real, authoritative and relevant to AI in transport.

Tell tale signs of AI generated content include repetitive phrasing, symmetrical paragraph structures and vague conclusions. Human written work typically demonstrates more selective emphasis and contextual judgement. Overall, AI tools are most effective as support mechanisms rather than substitutes for critical thinking. In ethical domains such as AI in transport, careful evaluation of sources and responsible interpretation of evidence remain essential.

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